

[DC] SharpView AR: Enhanced Visual Acuity for Out-of-Focus Virtual Content

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ABSTRACT

In augmented reality (AR), information is often distributed between real world and virtual contexts, and often appears at different distances from the user. Therefore, integrating information between real and virtual contexts raises the issues of (1) *context switching*, where users must switch visual and cognitive attention between information sources, (2) *focal distance switching*, where users must accommodate (change the shape of the eye's lens) to see, in sharp focus, information at a new distance, and (3) *transient focus blur*, where users observe retinal blur for a period of time. Switching context between sources of information from different focal distances is perceptually demanding and has performance consequences. Therefore, this research will leverage existing work in focus correction through image processing to develop and evaluate *SharpView fonts* for AR displays. A SharpView font is designed to look better (sharper) when seen out of focus than a regular font. The research hypothesizes that SharpView fonts will facilitate context switching and focal distance switching, and mitigate transient focus blur, resulting in measurably improved legibility and user comfort user performance for out-of-focus virtual content.

1 INTRODUCTION

When using optical see-through (OST) AR displays, interacting with virtual content requires the observer's eyes to focus on the display's optical depth. Furthermore, most current optical see-through (OST) AR displays place virtual information at a fixed focal distance (e.g., Microsoft HoloLens, Google Glass). In contrast, the real world is complex, and information appears at a range of different focal distances. As a result, in OST AR, integrating information between AR and the real world brings three AR interface design issues: context switching, focal distance switching, and transient focus blur. *Context switching* means switching visual and cognitive attention between real and virtual contents [5]. Additionally, rapidly shifting eye focus from one focal distance to another focal distance and vice versa is defined as *focal distance switching* [5]. During focal distance switching, users observe only one object (either real or virtual content) in focus. Other information becomes blurred for a period of time (around 360 milliseconds [4]), which generates a *transient focus blur* problem in AR. Previous research has also found that continuously shifting eye focus between distances and contexts significantly affects task performance, reduces comfort, and increases fatigue and eye strain [5, 6]. Further, focal distance switching has a strong correlation with two essential depth cues: accommodation and vergence. While observing virtual contents through an OST AR display, the accommodative depth cue is usually fixed by the display's optical focal distance. Therefore, a viewer can be required to visually fuse an object with a vergence demand quite different from its accommodative demand. This inconsistency between the depth cues is called the *accommodation-vergence mismatch problem* — it is a pervasive problem AR displays. It can cause eye strain, double

vision, or visual suppression of the image in one eye [6]. Although work on multi-focal, varifocal, light field, and holography AR displays [8, 11] promises to eventually mitigate this problem, currently most of these displays exist as workbench prototypes, and it is not known whether or how long it will take for these solutions to mature into wearable displays. Pre-filtering for focus manipulation has also been applied to the goal of allowing observers with refractive errors to view images without wearing their corrective optics. For example, Montalto et al. [9] developed a novel technique to generate imagery with improved sharpness for individuals with refractive vision problems, including myopia, hyperopia, presbyopia, and astigmatism. However, focus correction techniques through image processing developed for the visually impaired has not been tested or validated for viewing out-of-focus virtual content. To our knowledge, my advisor and colleagues were the first to examine the problem of AR content blurring, and proposed the SharpView algorithm for general images [7]. However, this work did not appropriately consider human visual aberration, and did not show promising results for short AR text labels and symbols. Xu et al. [12] proposed the first fully software-based visual aberration correction for users with refractive errors in VR environments. Therefore, finding an effective and appropriate image processing method to solve or mitigate the out-of-focus problem in AR is a novel and challenging research avenue.

2 RESEARCH GOALS

My dissertation research hypothesizes that SharpView fonts will facilitate context switching and focal distance switching, resulting in measurably improved legibility, user comfort, and user performance. To explore this hypothesis, I will further quantify the impact of context switching and focal distance switching on human performance. After that, my research goal is to develop, validate and tune the SharpView algorithm parameters optimized for OST AR, considering the focal disparity between real and AR information. Finally, in my Ph.D. research, I will quantitatively measure the degree to which SharpView improves human performance in AR applications.

3 CURRENT PROGRESS

3.1 Replicate and Extend AR Text Search Task

To explore AR display context switching and focal distance switching, we have replicated the text-based visual search task developed by Swan, Gabbard, and collaborators [5] on the AR haploscope [10]. This replication extends the previous work by including the variable *stereopsis* (stereo, mono), and by fully crossing the variables of context switching and focal distance switching [1, 2]. As described in my master's thesis [1], data from 24 participants replicates the same negative effects of context switching, focal distance switching, and transient focus blur. The many physical differences between the AR haploscope and the monocular display used in the previous work are consistent with the hypothesis that these findings broadly generalize to AR user interfaces. Although a paper reporting these results is currently being written, an abstract was published at IEEE Virtual Reality 2020 [2]. Other than Gabbard et al. [5] and our recent replication, we are not aware of any additional experiments that have

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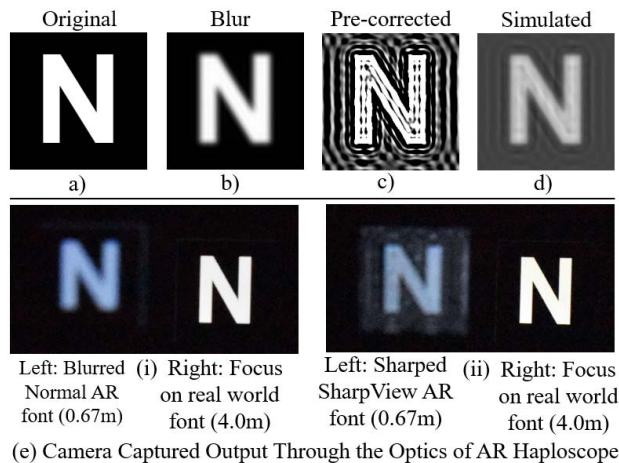


Figure 1: SharpView method with the single letter “N”. (a) Original image without any pre-correction. (b) Blurred image by Zernike PSF for 3.33-meter focal disparity. (c) Pre-corrected image, generated with a constrained total variation-based deconvolution method. (d) Simulated output showing improved visual acuity. (e) (i) Using the original font (a), when focused on the real world font at 4 meters, the AR font is blurred at 0.67 meters. (ii) Still focused on the real world font but using the pre-corrected font (c). The fonts in both (i) and (ii) are viewed out of focus, but (ii) looks better (sharper) than (i). The images in the bottom row (e) were photographed through the optics of a workbench-mounted AR Haploscope [10]. Camera parameters: Nikon D3400, aperture size: 3mm.

systematically examined both context and focal distance switching in the same experiment.

3.2 Develop and Validate SharpView Algorithm

In previous image processing work for focus correction, generating a pre-corrected image requires specifying parameters, mainly the point spread function of a visually impaired eye and the viewing distance. However, these methods have not yet been optimized for OST AR, where parameter trade-offs may be completely different. Considering the previous image processing work for focus correction, along with my advisor and collaborator Alexander Plopski, I have determined that the method described by Montalto et al. [9] is the most promising. Following the Montalto et al. [9] method, for OST AR, I modeled the visual aberration (see figure 1b) in terms of the second-order *Zernike polynomials* by considering these parameters: (1) pupil diameter, (2) the focal disparity between the real world and AR contexts, (3) the absolute distance to the real context, (4) the focal distance to the AR context, and (5) the focused object size. We used a constrained total variation deconvolution algorithm [3]. In addition, we adapted the relative total variation term from Montalto et al. [9] to systematically control the ringing and contrast in the pre-corrected AR image (see figure 1c). Our simulated image shows sharper and improved visual acuity compared to the blurred image (see figure 1d). After that, I used a camera-based experiment to determine whether this method shows any optical improvement through the AR Haploscope at focal distance differences of 3.33 meters (real font at 4 meter and AR font at 0.67 meter). When we displayed the normal AR font at 0.67m, the camera captured blurred images (see figure 1e-(i)). However, instead of a normal AR font when we displayed our simulated pre-corrected image, a visually sharp and more legible image was captured by the camera (figure 1e-(ii)).

4 DISCUSSION

Although context switching, focal distance switching, and transient focal blur in AR are very common, only a few research efforts

have explored these issues. In addition, improving the clarity of out-of-focus graphical content in AR is still an open question. If my dissertation research is successful, SharpView fonts will prove to be a novel AR capability, which will improve the problem of viewing AR content in conditions of focal disparity. However, even if SharpView fonts fail to live up to this promise, as a minimal contribution my dissertation project will contribute a deeper understanding of the effects of AR display context switching and focal distance switching. In the future, I want to generalize this method for different types of fonts, short text labels, and symbols in AR. My initial goal is to conduct a camera-based experiment to determine whether this method shows any notable improvement at other focal distance differences. After that, my main goal will be to verify the algorithm parameters with human participants. I welcome constructive feedback and suggestions on these ongoing and future lines of research.

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